VRYFD

Setup & Run Guide

Complete step-by-step instructions to set up, compile, deploy, and run the VRYFD decentralized attendance system with **$VRYF token rewards** on your machine.

Layer	Technology	Purpose
Smart Contracts	Solidity + Hardhat + OpenZeppelin	ERC-1155 attendance, badges, rewards
$VRYF Token	ERC-20 (custom)	Reward token: 10/25/100/250/500 per action
Frontend	Next.js + wagmi + RainbowKit	Wallet connect, ENS, UI
Backend	Express + ethers.js	IPFS relay, certificates, QR codes
Blockchain	Ethereum Sepolia Testnet	Free test transactions
Storage	IPFS via Pinata	Badge images, metadata, certificates

$VRYF Token Reward Structure

Action	$VRYF Earned
Check in to an event	+10 VRYF
Claim a POAP badge	+25 VRYF
Bronze Explorer (5 events)	+100 VRYF
Silver Pioneer (10 events)	+250 VRYF
Gold Legend (20 events)	+500 VRYF
Max supply cap	1,000,000 VRYF

1. Prerequisites

Node.js 18+ — Download from https://nodejs.org — run **node --version** to verify

Git — Download from https://git-scm.com

VS Code — Download from https://code.visualstudio.com

MetaMask — Browser extension from https://metamask.io — create a **separate VRYFD Dev account**

Sepolia ETH — You already have Sepolia tokens. If you need more: https://sepoliafaucet.com

Pinata Account — Sign up at https://pinata.cloud for IPFS storage (free tier works)

WalletConnect ID — Create project at https://cloud.walletconnect.com (free)

Etherscan API Key — Sign up at https://etherscan.io → API Keys → Create

2. Unzip & Open in VS Code

Step 2.1 — Unzip the project

Unzip **vryfd-project-sepolia.zip** to any folder on your computer (Desktop, Documents, etc.).

Step 2.2 — Open workspace in VS Code

Open VS Code, then go to **File** → **Open Workspace from File** and select **vryfd.code-workspace** inside the unzipped folder. This loads the multi-root workspace with separate folders for Smart Contracts, Frontend, and Backend.

Step 2.3 — Install recommended extensions

VS Code will show a popup asking to install recommended extensions (Solidity, Prettier, Tailwind CSS, ESLint). Click **Install All**.

3. Environment Setup

Step 3.1 — Create .env file

Open the VS Code terminal (Ctrl+`) and run:

```
cp .env.example .env
```

Then open **.env** and fill in these values:

Variable	Where to Get It

`PRIVATE_KEY`	MetaMask → VRYFD Dev account → Account Details → Show Private Key
`SEPOLIA_RPC`	https://rpc.sepolia.org (default, works out of the box)
`ETHERSCAN_API_KEY`	https://etherscan.io → Sign Up → API Keys → Create
`PINATA_JWT`	Pinata Dashboard → API Keys → New Key → copy JWT
`NEXT_PUBLIC_WALLETCONNECT_PROJECT_ID`	https://cloud.walletconnect.com → New Project → copy ID

WARNING: Never commit the .env file to Git. It contains your private key! Use a separate VRYFD Dev MetaMask account, never your real wallet.

4. Install Dependencies

Run this single command from the project root:

```
npm run install:all
```

This installs dependencies for all three parts: root (Hardhat + Solidity), frontend (Next.js + wagmi), and backend (Express + ethers). Takes 1–3 minutes.

5. Compile & Test Smart Contracts

Step 5.1 — Compile

```
npx hardhat compile
```

This compiles both **VRYFToken.sol** and **VryfdAttendance.sol** and generates ABIs in artifacts/.

Step 5.2 — Run tests

```
npx hardhat test
```

You should see **18 passing tests** covering: $VRYF token basics, check-in earns 10 VRYF, badge claim earns 25 VRYF, milestone earns 100/250/500 VRYF, soulbound transfer restriction, certificates, and admin functions.

Step 5.3 — Copy ABI to frontend & backend

```
cp artifacts/contracts/VryfdAttendance.sol/VryfdAttendance.json
frontend/src/abis/
mkdir -p backend/artifacts
cp artifacts/contracts/VryfdAttendance.sol/VryfdAttendance.json
backend/artifacts/
```

6. Deploy to Sepolia Testnet

Make sure your VRYFD Dev wallet has Sepolia ETH, then run:

```
npx hardhat run scripts/deploy.js --network sepolia
```

The deploy script automatically:

1. Deploys **VRYFToken** (your $VRYF ERC-20 token)
2. Deploys **VryfdAttendance** (linked to the token)
3. Grants **MINTER_ROLE** so attendance actions auto-mint $VRYF

4. Verifies both contracts on **Etherscan**

Copy the two addresses printed by the script into your **.env** file:

```
NEXT_PUBLIC_CONTRACT_ADDRESS=0x_attendance_address_here
NEXT_PUBLIC_VRYF_TOKEN_ADDRESS=0x_vryf_token_address_here
```

TIP: After deployment, view your contracts at https://sepolia.etherscan.io. Users can add $VRYF to MetaMask by clicking Import Token and pasting the VRYF Token address.

7. Run the Application

You need **two terminal windows** running simultaneously:

Terminal 1 — Start the frontend

```
cd frontend
npm run dev
```

Opens at **http://localhost:3000** — this is the main UI users interact with.

Terminal 2 — Start the backend

```
cd backend
npm run dev
```

Opens at **http://localhost:3001** — handles IPFS pinning, gasless relaying, certificate generation, and the verification API.

What you should see:

Action	What Happens	Page
Connect wallet	MetaMask popup → approve → ENS name + avatar shown	All pages
Click an event	Check-in panel with EIP-712 sign flow	Events
Sign message	Gasless signature sent to backend relay queue	Events
Claim badge	Soulbound POAP minted → visible in MetaMask	Events
Check $VRYF balance	Token balance updates: +10 check-in, +25 badge	Profile
View badges	Gallery of earned POAPs + milestone progress bars	Badges
Leaderboard	Top 10 ENS names ranked by attendance + $VRYF score	Leaderboard
Certificate	Generated image with QR code, pinned to IPFS	Certificates
Scan QR	Opens verification portal with blockchain proof	External

8. Sepolia Network & MetaMask Setup

Sepolia network details:

Setting	Value
Network Name	Sepolia
Chain ID	11155111
RPC URL	https://rpc.sepolia.org
Currency Symbol	ETH
Block Explorer	https://sepolia.etherscan.io

How to create the VRYFD Dev account in MetaMask:

1. Open MetaMask in your browser
2. Click the account circle (top-right) → **Add account or hardware wallet**
3. Choose **Add a new account**
4. Name it **VRYFD Dev** and click Create
5. Switch MetaMask to **Sepolia** network

How to export the private key:

1. Make sure **VRYFD Dev** is selected in MetaMask
2. Click the three dots menu (top-right of the account)
3. Click **Account details**
4. Click **Show private key** and enter your MetaMask password
5. Copy the key (starts with 0x...) into your .env file as PRIVATE_KEY

> **WARNING:** Never use a wallet with real funds. The VRYFD Dev account is a throwaway wallet for development only. Sepolia ETH has zero real-world value.

How to add $VRYF token to MetaMask:

1. After deployment, copy the **VRYF Token address** from the terminal output
2. In MetaMask, click **Import tokens** at the bottom of the assets list
3. Paste the VRYF Token address — Symbol and decimals auto-fill
4. Click **Add Custom Token** → **Import**

5. Your $VRYF balance will now show in MetaMask alongside your Sepolia ETH

9. VS Code Shortcuts & Tasks

The workspace includes pre-configured tasks and debug configs. Access them via:

Shortcut	What It Does
`Ctrl+Shift+P → "Run Task"`	Shows all project tasks (compile, test, deploy, start servers)
`F5`	Launch debugger (pick Next.js, Backend, or Hardhat test config)
`Ctrl+Shift+B`	Run the default build task (compile contracts)
`Ctrl+``	Toggle integrated terminal
`Ctrl+Shift+E`	Switch between workspace folders (Contracts / Frontend / Backend)

10. Troubleshooting

'npx hardhat' not found

Make sure you ran **npm install** in the project root first.

Contract deployment fails

Check that your VRYFD Dev wallet has Sepolia ETH and your PRIVATE_KEY is correct in .env. The deploy needs ~0.01 Sepolia ETH for gas.

Frontend can't connect to contract

Verify NEXT_PUBLIC_CONTRACT_ADDRESS in .env matches the deployed address. **Restart the dev server** after changing .env (Ctrl+C, then npm run dev again).

$VRYF balance not showing

Make sure NEXT_PUBLIC_VRYF_TOKEN_ADDRESS is set in .env. In MetaMask, use Import Token with the VRYF contract address.

IPFS upload fails

Check your PINATA_JWT in .env. Make sure you created an API key with pinning permissions.

ENS name not resolving

ENS lives on Ethereum mainnet. Your wagmi config includes mainnet for ENS resolution — make sure you have an Alchemy API key set in NEXT_PUBLIC_ALCHEMY_API_KEY.

MetaMask shows wrong network

Switch MetaMask to **Sepolia** (Chain ID: 11155111). RainbowKit will prompt users to switch automatically if they're on the wrong network.

Tests fail after contract changes

Run **npx hardhat clean** then **npx hardhat compile** before re-running tests.

Deploy says 'insufficient funds'

Your VRYFD Dev account needs Sepolia ETH. Get more from https://sepoliafaucet.com or https://www.alchemy.com/faucets/ethereum-sepolia.

11. Project Structure

Folder / File	Owner	Purpose
`contracts/VRYFToken.sol`	Person A	$VRYF ERC-20 token with minting roles
`contracts/VryfdAttendance.sol`	Person A	ERC-1155 attendance, badges, milestones, certificates
`scripts/deploy.js`	Person A	Deploys both contracts, links them, grants MINTER_ROLE
`test/VryfdAttendance.test.js`	Person A	18 tests covering all features + token rewards
`frontend/src/components/`	Person B	CheckInCard, Leaderboard, WalletConnect components
`frontend/src/hooks/`	Person B	useENS, useContract, useVRYFToken hooks
`backend/src/services/`	Person C	IPFS pinning, gasless relay, certificate generation
`backend/src/routes/`	Person C	API endpoints: /ipfs, /relay, /certificates, /verify

12. API Endpoints

Method	Endpoint	Description
POST	`/api/ipfs/upload-image`	Upload badge image to IPFS
POST	`/api/ipfs/badge-metadata`	Create & pin POAP metadata
POST	`/api/relay/checkin`	Submit gasless check-in signature
POST	`/api/relay/flush`	Force-submit batch to chain
GET	`/api/relay/status`	Check relay queue length
POST	`/api/certificates/generate`	Generate certificate + QR + IPFS
GET	`/api/verify/:id`	Public certificate verification
GET	`/api/health`	Server health check